Problem 10: dictionary | [dik-shuh-ner-ee] | noun
13 Points

Problem ID: `dictionary`
Rank: `4`

Introduction

Big Ben, as we know from an earlier contest, is a huge fan of Frieren! After watching the anime an infinite number of times, he's going to start reading the manga. He came back to Berkeley after taking the BART Red Line and the 38R Geary to Kinokuniya Bookstore in San Francisco's Japantown and picking up the first book. He just prepared a cup of hot chocolate and is on his couch with the fluffiest blanket he could find. After opening the first page, he's come across a bit of an issue: he doesn't know any words! Luckily, he has a dictionary.



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Problem Statement

Big Ben's dictionary contains definitions for $\mathbf{N}$ distinct words; the words are numbered 1 through $\mathbf{N}$. The i-th word has a definition which starts with the word $\mathbf{A}_i$ and has an associated value of $\mathbf{V}_i$. Big Ben's starting happiness is $h = 0$.

When he starts reading the definition of the i-th word, his happiness h will become $h \oplus \mathbf{V}_i$ (where $\oplus$ represents the bitwise XOR). Then, as he reads this definition, he will come across the word $\mathbf{A}_i$. He'll try to find the definition of the $\mathbf{A}_i$-th word and repeat the process above (this time with the $\mathbf{A}_i$-th word), or be satisfied with the recursion of definitions and stop there. For every integer i from 1 to $\mathbf{N}$, Big Ben wants you to figure out, the maximum happiness h_i he can end up with at the optimal stopping point, given that he starts by looking at the i-th word.

Input Format

The first line of the input contains an integer $\mathbf{T}$ denoting the number of test cases that follow. For each test case:

- The first line contains a single integer $\mathbf{N}$ denoting the number of words in the dictionary.
- The second line contains $\mathbf{N}$ space-separated integers $\mathbf{A}_1\ \mathbf{A}_2\ \ldots\ \mathbf{A}_N$, where $\mathbf{A}_i$ denotes the first word in the definition of word i.
- The third line contains $\mathbf{N}$ space-separated integers $\mathbf{V}_1\ \mathbf{V}_2\ \ldots\ \mathbf{V}_N$, where $\mathbf{V}_i$ denotes the value associated with the i-th word.

Output Format

For each test case, output a single line containing $\mathbf{N}$ space-separated integers $h_1\, h_2\, \ldots\, h_{\mathbf{N}}$. The i-th integer h_i represents the maximum happiness Big Ben can have if he starts at word i.

Constraints

Time limit: **2 seconds**.
$1 \leq \mathbf{T} \leq 100$
$1 \leq \mathbf{N} \leq 2 \cdot 10^5$
$1 \leq \mathbf{A}_i \leq N$
$0 \leq \mathbf{V}_i < 2^{30}$
The sum of $\mathbf{N}$ across all test cases in a test file does not exceed $2 \cdot 10^5$.

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Sample Test Cases

Sample Input Download

```
4
2
1 2
1 2
2
2 1
3 7
2
1 1
3 7
8
6 8 7 3 6 7 8 4
6 8 7 3 6 7 8 4
```

Sample Output Download

```
1 2
7 7
3 7
14 15 15 12 14 15 15 15
```

Sample Explanations

For the first test case: there are two words in Big Ben's dictionary. The first word's definition contains itself, and the second word's definition also contains itself. The first word has an associated happiness value of 1, while the second word has an associated happiness value of 2. Suppose Big Ben starts with the first word. His happiness starts at $h = 0$. Then, he opens the dictionary to the first word and his happiness becomes $h = 0 \oplus \mathbf{V}_1 = 1$. His optimal choice is to stop at this moment, which means that $h_1 = 1$. Suppose Big Ben starts with the second word. His happiness likewise starts at $h = 0$. Then, he opens the dictionary to the second word and his happiness becomes $h = 0 \oplus \mathbf{V}_2 = 2$. His optimal choice is to stop at this moment, which means that $h_2 = 2$.

For the second test case: there are two words in Big Ben's dictionary. If Big Ben starts with the first word, his optimal choice is to open the dictionary to the first word, then the second word, then the first word, and then stop. Thus, $h_1 = \mathbf{V}_1 \oplus \mathbf{V}_2 \oplus \mathbf{V}_1 = 3 \oplus 7 \oplus 3 = 7$. If Big Ben starts with the second word, his optimal choice is to open the dictionary to the second word, and then stop. Thus, $h_2 = \mathbf{V}_2 = 7$.

For the third test case: there are two words in Big Ben's dictionary. If Big Ben starts with the first word, his optimal choice is to open the dictionary to the first word, and then stop. Thus, $h_1 = \mathbf{V}_1 = 3$. If Big Ben starts with the second word, his optimal choice is to open the dictionary to the second word, and then stop. Thus, $h_2 = \mathbf{V}_2 = 7$.

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